Asignación 3

Tuesday, April 2, 2024 12:23 PM

1. Consider the following symbol frequencies for a five-symbol alphabet:

Symbol	Frequency
A	.32
B	.25
C	[illegible]
D	[illegible]
E	[illegible]

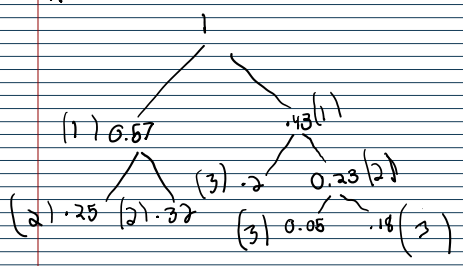
What is the average encoding length of an optimal prefix-free code? *[5pts]*

2. Consider the following symbol frequencies for a five-symbol alphabet:

Symbol	Frequency
A	[illegible]
B	[illegible]
C	.35
D	[illegible]
E	[illegible]

What is the average encoding length of an optimal prefix-free code? *[5pts]*

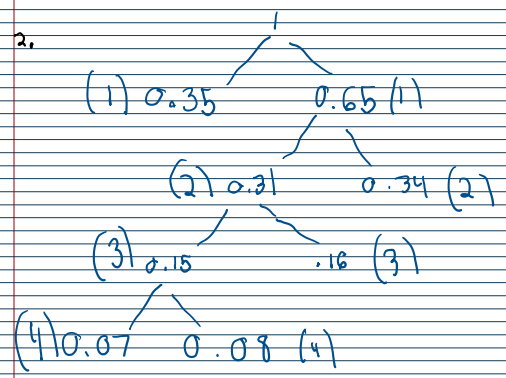
1.



$0.57 + 0.43 + (2).25 + .32(2) + (2).2 + 0.23(2) + 0.05(3) + 0.18(3)$

Average $- 3.69$

2.

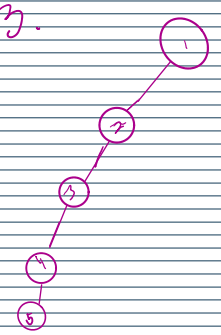


$0.35 + 0.65 + 2(0.31) + 0.34(2) + 0.15(3) + 0.16(3) + (4)0.07 + 0.08(4)$

Average $- 3.13$

3. In terms of *n*, where *n* = |Σ| denotes the alphabet size, what is the maximum number of bits that Huffman's greedy algorithm might use to encode a single symbol? Explain your answer. *[5pts]*
Hint: Think of the different types of trees that could result.

3.



n = elementos

n − 1

The worst case will be n − 1